PUVIRAJ R

Data & AI Engineer | Python | SQL | FastAPI | React | Power BI | Enterprise SaaS

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PROFESSIONAL SUMMARY

Data & AI Engineer with experience building enterprise automation platforms, AI-assisted workflow systems, and scalable data pipelines for Smart Buildings and IoT. Experienced in Python, SQL, FastAPI, React, PostgreSQL, and enterprise data engineering with a strong focus on workflow automation and AI integration.

TECHNICAL SKILLS

- Programming: Python, SQL, TypeScript, JavaScript
- Backend: FastAPI, REST APIs, SQLAlchemy
- Frontend: React, Tailwind CSS, Vite
- AI & Automation: AI Agent Integration (OpenAI), Function Calling, Prompt Engineering, Rule-Based Fuzzy Matching (RapidFuzz)
- Database: PostgreSQL, MySQL
- Data & BI: Power BI, Excel, Pandas, ETL, Data Validation & Cleaning
- DevOps: Docker, Git, GitHub
- Domain: Smart Buildings, Internet of Things (IoT), Building Management Systems (BMS), Computer-Aided Facility Management (CAFM), BACnet Protocol, Energy Analytics

PROFESSIONAL EXPERIENCE

Associate Energy Analyst *Jul 2025 – Present*

Nectar Building Technologies Pvt Ltd — Coimbatore, Tamilnadu

- Built a 5-tier fuzzy-matching engine using Python, Pandas, and RapidFuzz to standardize BMS/BACnet equipment and point data across multi-site enterprise portfolios. Processed 70,000+ data points in a single portfolio analysis using domain-specific validation rules to resolve polarity mismatches, compound-tag containment, abbreviation collisions, and other vendor-specific naming inconsistencies
- Delivered end-to-end IoT and BMS onboarding across 100+ sites for 10+ enterprise clients, reducing manual onboarding effort by 60% through Python-based automation processing 100,000+ data points per batch.
- Developed structured ETL workflows for asset mapping, space hierarchy validation, and IoT sensor reconciliation, producing analytics-ready datasets for Power BI dashboards and operational KPI reporting
- Led a 6-member analytics team, coordinating sprint planning, mentoring team members, and collaborating with backend developers, facility teams, vendors, and enterprise clients to deliver projects on schedule.
- Completed a 3-month onsite engagement in Dubai, UAE, working directly with enterprise clients and facility operations teams to support IoT/BMS onboarding, data validation, and deployment activities.

Relationship Manager *May 2023 – Jun 2025*

PNB MetLife India Insurance Co Ltd- Coimbatore, Tamilnadu

- Managed a portfolio of 10,000+ customer accounts; applied behavioural trend analysis and data segmentation to prioritise outreach, contributing to measurable improvement in cross-sell conversion and policy renewal rates.
- Produced structured performance reports and business analysis summaries to support operational decision-making and branch-level planning.

PROJECTS

BMS Point Mapping Automation Tool

(Python , Pandas ,RapidFuzz ,OpenPyXL,OpenAI)

- Designed and developed a scalable BMS data standardization engine using a 5-tier matching algorithm and a 368-pair alias dictionary to normalize vendor-specific point names across enterprise building portfolios.
- Engineered domain-specific validation rules to resolve short-code collisions, reverse-containment patterns, and instance-specific false matches, improving matching accuracy across diverse client datasets
- Processes over 100,000 data points in under two minutes and, during benchmarking, identified approximately 16% incorrect matches in an LLM-generated mapping of the same dataset through rule-based validation.

AI-Powered Project Management Platform

(Python , FastAPI , React , PostgreSQL , Docker , AI Agents)

- Designed and developed an AI-powered project and workflow management platform that streamlines project planning, task management, document management, analytics, and AI-assisted workflow automation in a single application.
- Implemented Excel-based bulk project creation, allowing users to upload a standardized project template that automatically generates projects, milestones, tasks, assignees, priorities, and deadlines—eliminating repetitive manual project setup.
- Integrated a function-calling AI agent using the OpenAI API, enabling users to create, update, assign, delete, and retrieve tasks using natural language, with an extensible architecture for future AI-driven workflows.
- Built a modular FastAPI backend, React dashboard, PostgreSQL database, and Docker-based architecture with REST APIs, role-based access control, real-time dashboards, and analytics for enterprise project management.

Daily Activity Tracker

(Flutter, SQLite, Excel, PDF)

- Architected and implemented an offline-first daily activity tracking application using Flutter, leveraging AI-assisted development while owning the application design, testing, debugging, and feature integration.
- Designed automated Excel and PDF report generation, reducing manual reporting effort by approximately 40%.

Global Climate & Energy Dashboard

(Power BI , DAX , Looker Studio)

- Designed and developed interactive Power BI dashboards with drill-down analysis, KPI visualization, and dynamic filtering for climate and energy datasets.
- Built calculated measures and KPIs using DAX to analyze energy consumption, emissions, and sustainability metrics.
- Published the project as a portfolio showcase demonstrating business intelligence, data modeling, and dashboard design capabilities.

EDUCATION

- MBA – HR & Business Analytics, Tamilnadu College of Engineering *2023-2025*
- B.E. – Mechanical Engineering, Tamilnadu College of Engineering *2017-2021*

LANGUAGES

English, Tamil, Malayalam

CERTIFICATIONS

- SQL – MySQL for Data Analytics and Business Intelligence Click Here
- Business Analysis Fundamentals — IIBA-endorsed Click Here

Portfolio Link : Click Here